

COSC 39: Theory of Computation

Problem 1. Write down the decision version of each of the following problems, then describe a polynomialtime self-reduction assuming an oracle to the decision problem.

1.

FINDMAXCLIQUE

- **Input:** An undirected graph G .
- **Output:** Compute a maximum-size clique in G .

2.

FIND3COLORING

- **Input:** An undirected graph G .
- **Output:** Compute a proper vertex 3-coloring of G (if exists).

Solution.

1.

FINDMAXCLIQUE

- **Input:** An undirected graph G .
- **Output:** Is there a size of clique k in G ?

$k = |V|, k - 1, \dots, 1$ choose k